

A closer look at the innovative PAGED-B score

To the Editor:

I wish to express my thoughts regarding the recent publication by Chun *et al.*¹ wherein they developed a new hepatocellular carcinoma (HCC) risk prediction score (PAGED-B score) in non-cirrhotic treatment-naïve adult patients who started antiviral therapy upon entering into hepatitis B e antigen (HBeAg)-positive chronic hepatitis B from chronic infection. The score was found to have high accuracy for the prediction of 5-year risk of HCC. While I appreciate the value of this rigorous study, some constructive suggestions that might enrich the authors' findings for the readership are offered, encouraging further exploration in this pivotal domain.

First, the consistency of model comparisons. This work employed time-dependent AUROC and Harrell's C-index to assess and compare the predictive ability of the PAGED-B score to current HCC risk prediction scoring models (PAGE-B, mPAGE-B, CAMD, and REAL-B). Nevertheless, I advocate incorporating consistency analysis such as correlation analysis like the Pearson correlation coefficient since it can evaluate the strength of the association between the PAGED-B score and the other four scoring systems, which may better validate whether the newly developed PAGED-B score offers comparable or superior risk prediction capabilities than established scores, and reveal differences in the scoring systems.² Consistency analysis may disclose variations and consistency levels in prediction outcomes among models, and give a more thorough assessment to enhance research accuracy.

Furthermore, although exclusion criteria establish baseline conditions for the study, such as a minimum threshold for serum HBV DNA levels and the exclusion of patients with severe active hepatitis, to ensure that the study participants have a certain initial viral load and degree of active hepatitis, these standards do not account for differences in antiviral treatment between patients. The research did not explain the consistency of treatment plans, such as treatment length and dose, treatment strategies for patients with varying treatment responses, medication resistance, adverse effects, and so on. Variances in therapy may result in variances in HCC risk across individuals

on the same treatment regimen, influencing the PAGED-B score.^{3,4} Future studies may include analysis of treatment regimen data by providing the percentage of patients with different alanine aminotransferase levels at treatment initiation to optimize research.

Finally, as an emerging tool for predicting HCC risk, the PAGED-B scoring model's clinical application value is determined not only by its predictive accuracy but also by the practical advantages that the model may provide to clinical decision-makers. Hence, I recommend adopting clinical decision curve analysis,⁵ which can visually demonstrate the potential benefits and risks of using the PAGED-B score for screening and treatment decision-making at various risk acceptance levels, as well as providing clinical doctors with clearer guidance on using the model more effectively in clinical practice.

In conclusion, Chun *et al.* used large-scale cohort analysis to add the moderate HBV DNA level into the current PAGE-B model and presented the PAGED-B scoring system to better predict the risk of HCC in patients with HBeAg-positive chronic hepatitis B. This is a significant achievement in the realm of customized healthcare, allowing clinicians to better identify high-risk patients and adapt monitoring and treatment techniques appropriately. My suggestion is only to further improve an already excellent study. I look forward to more of the authors' insightful works in the future.

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Supplementary data

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Author names in bold designate shared co-first authorship

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